

⇒ Rocket Trajectory Sim. Exercise 1 Notes:

Formulate a 1D model for rocket motion in the absence of outside forces. Using Euler method for numerical integration, compute the velocity of a V2 rocket & compare it with the velocity computed using Tsiolkovsky's rocket equation.

- > Specific impulse is a measure of how efficiently a rocket engine generates thrust from propellant, & is given by

$$I_{sp} = \frac{T}{\dot{m} g_0} \text{ [s]}$$

where

T = rocket thrust [N]

$\dot{m}$ = propellant mass flow rate [kg/s]

g_0 = standard gravity = 9.81 [m/s²]

Rearranging & solving for thrust, we obtain

$$T = I_{sp} \dot{m} g_0$$

Effective exhaust velocity is given by

$$v_e = I_{sp} g_0$$

Thus, thrust can be expressed as

$$T = v_e \dot{m}$$

- > Formulate the 1D model for rocket motion using Newton's 2nd Law:

$$F = \frac{d}{dt}(mv) = m \frac{dv}{dt} \rightarrow T = m \frac{dv}{dt}$$

$$\frac{dv}{dt} = \frac{T}{m} = \frac{v_e \dot{m}}{m_{\text{empty}} + m_{\text{propellant}} - \dot{m} t}$$

$$\frac{dm}{dt} = -\dot{m}$$

where $m_{\text{empty}} = m_{\text{total}} - m_{\text{propellant}}$

> Can solve 1D equation of motion using Euler's method

Euler's method for integrating ODEs:

Consider the function

$$\frac{dv}{dt} = f(t, v)$$

First, approximate the derivative of v using a forward finite difference

$$\left. \frac{dv}{dt} \right|_{t_0} \approx \frac{v(t_0 + \Delta t) - v(t_0)}{\Delta t}$$

Then, rearrange to solve for v at the next time step

$$v(t_0 + \Delta t) \approx \left. \frac{dv}{dt} \right|_{t_0} \Delta t + v(t_0)$$

One can integrate the solution:

- Start with an initial value
- Compute the derivative
- Find the next value

Example:

$$V_{i+1} = \left. \frac{dv}{dt} \right|_{t_i} \Delta t + V_i$$

for i in $[0, n]$, with n being the final index

> Compare numerical solution of velocity to that obtained from the Tsiolkovsky rocket equation, given by

$$\Delta V = v_e \ln\left(\frac{m_i}{m_f}\right)$$

where

m_i = initial rocket mass

m_f = final rocket mass = $m_{\text{empty}} + m_{\text{propellant}} - \dot{m}t$

(For derivation of Tsiolkovsky rocket equation, see "Orbital Dynamics Part 48" YouTube video by Orbital Dynamics LA)

> V2 rocket properties:

Empty mass: $M_{\text{empty}} = 4000 \text{ [kg]}$

Propellant mass: $M_{\text{propellant}} = 8800 \text{ [kg]}$

Total mass: $M_{\text{total}} = M_{\text{empty}} + M_{\text{propellant}} = 12800 \text{ [kg]}$

Propellant mass flow rate: $\dot{m} = 129.4 \text{ [kg/s]}$

Effective exhaust velocity: $V_e = 2050 \text{ [m/s]}$